

1. Data Acquisition

DataBento MBP-10
6 symbols $\times$ 4 days
~7M tick events / symbol

2. Offline Feature Research

IC / ICIR scoring per feature
Greedy conditional IC selection
Outputs selected_features.yaml

⋮
optional: reduced
feature config

3. Data Preparation

FeaturePipeline.fit on train only
Welford online z-score normalisation
18 memmap files (6 symbols $\times$ 3 days)
val + test DataFrames

4. Config Guardrails

50 pre-flight checks
FATAL $\rightarrow$ ValueError / abort
WARN $\rightarrow$ prompt or auto-proceed

pass

5. RL Training

Algorithms: TD3 · DDPG · PPO · Random
StreamingTradingEnvXY episode shuffle
DSR or log-return reward
Runtime health monitor (early stop)
Checkpoints + MLflow logging

6. Post-Training Evaluation

Deterministic rollout on val + test
Sharpe · Sortino · drawdown metrics
Buy-and-hold / TWAP / VWAP benchmarks
Statistical significance tests
Permutation + IG explainability